

performance on the GOAL dimension. However, we remind readers that LLMs are known to have biases and problems for evaluation, including positional bias (Wang et al., 2023), factual inconsistency (Luo et al., 2023), favoring native speakers (Liang et al., 2023). Therefore, one should be aware of the influence of these potential biases when interpreting our results. Future versions of SOTOPIA-EVAL may further improve LLM-based evaluation quality using recent methods, such as involving multiple LLMs Chan et al. (2023) and training larger LLM evaluators Zhang et al. (2023).

6 EVALUATING SOCIAL INTERACTION BETWEEN LLMs IN SOTOPIA

We analyze models’ interactions and performance on SOTOPIA to understand their social intelligence. Table 2 presents the models’ average scores when interacting with different *partner models* (i.e., the model it is paired with in interaction, Fu et al. 2023; Hu et al. 2020).⁹ GPT-4 performs best on most dimensions, followed by GPT-3.5, Llama-2-70b-chat, and MPT-30b-chat.

Different trends from static

benchmarks Llama-2-70b-chat has relatively low scores in all dimensions compared to GPT-3.5 (except when MPT-30b-chat is the reference model, which is likely due to the fact that MPT-30b-chat is a much weaker model compared to other models in our experiments). This finding diverges from various static language understanding benchmarks showing that Llama-2-70b-chat is on par or better than GPT-3.5 (Li et al., 2023b; Touvron et al., 2023; Liang et al., 2022).¹⁰ We hypothesize that this is because Llama-2-70b-chat is less heavily trained on human feedback/user interaction data than GPT-3.5.

Through inspecting the interactions between Llama-2-70b-chat (MPT-30b-chat) and other models, we find that Llama-2-70b-chat and MPT-30b-chat often struggle to maintain their persona (Figure H.3), move the conversation forward (Figure H.4), and respond to the other agent actively (Figure H.5). Performing well on static benchmarks does not guarantee success in interactive scenarios, thus highlighting the importance of dynamic benchmarks like SOTOPIA-EVAL (Lee et al., 2023).

G-4	3.32	2.70	2.42	1.76
G-3.5	2.80	2.29	2.09	1.41
L-2	2.69	2.28	1.88	1.45
MPT	2.68	2.33	2.03	1.28
	G-4	G-3.5	L-2	MPT

Figure 3: Pairwise overall performance of models. G-4/G-3.5/L-2 denote GPT-4/GPT-3.5/Llama-2-70b-chat.

Weaker partners models weaken their conversation partners Figure 3, shows the overall performance of model pairs, which is the average performance across different dimensions. It is noticeable that a reference model that under-performs in SOTOPIA can lead to worse performance of other models.

For example, in a scenario where agents try to find a mutual friend (Figure H.6). The task fails for both GPT-4 and Llama-2-70b-chat because Llama-2-70b-chat consistently fails to answer the previous question even after GPT-4 attempts to steer the conversation back to the right track (e.g., “I noticed you didn’t answer my question about whether you know my friends or not.”). Since most of our social scenarios are fundamentally cooperative, the collapse of communication could be due to models’ lack of “cooperation” abilities (Oudouard & Price, 2023).

All models are at risk of divulging secrets and violating norms Table 2 shows that all models have a negative score in the SOC and SEC dimensions. Even though GPT-4 performs better in most dimensions, it is not better than other models in the SOC and SEC dimensions. For example, in a scenario where one needs to persuade a close friend to confess, the model leaks their secret at the beginning of the conversation (Figure H.7). This further shows the importance of considering multiple dimensions when evaluating models’ social intelligence.

	Dim.	Range	GPT-4	GPT-3.5	Llama-2	MPT
	SOC	[-10, 0]	-0.07	-0.08	-0.11	-0.09
	SEC	[-10, 0]	-0.14	-0.08	-0.14	-0.07
	FIN	[-5, 5]	0.81	0.46	0.40	0.28
	REL	[-5, 5]	1.94	1.23	0.91	0.58
	KNO	[0, 10]	3.73	3.40	3.11	2.11
	GOAL	[0, 10]	7.62	6.45	5.38	4.10
	BEL	[0, 10]	9.28	9.15	8.10	6.17

Table 2: The aggregated performance of each model by averaging across different partner models. The best performance for each dimension is bolded when significantly better than the second best in t-test ($p < 0.05$).

⁹Presented are automated evaluation results. The human evaluation shows a similar trend, see Table G.3

¹⁰Some reported results could come from different versions of GPT-3.5.

Models sometimes use creative strategies to accomplish goals We also find that models, especially GPT-4, could come up with “out-of-the-box” solutions to social problems. For example, when the agent is asked to take turns driving on the road trip, the agent (i.e., GPT-4), instead of directly rejecting their friend’s request, proposes “How about we pull over for a bit and get some rest?” (Figure H.8). Additionally, in the scenario where two agents make a plan to improve the company’s financial status, agents figure out strategies such as “having a small group tasked with identifying potential suppliers”, “while we conduct the search for new suppliers, we continue to negotiate with our current supplier” (Figure H.9).

7 DIFFERENCES BETWEEN MODELS AND HUMANS IN SOCIAL INTERACTION

To understand how humans and models interact differently in SOTOPIA, we conduct a study where humans interact with models or each other under this role-playing setting (§2). Specifically, we build a chat interface that allows humans and models to interact with each other in a turn-based manner.

To fully see the difference between humans and models, we select the most challenging scenarios following Dennis et al. (2020); Swayamdipta et al. (2020). Specifically, we consider the gap between the estimated maximum rewards (average reward plus three standard deviations) of all models and the estimated minimal rewards (average reward minus three standard deviations) of the target model as the difficulty of the task for the model. All maximum and minimum rewards are bounded by the corresponding range. Estimating maximum and minimum rewards with standard deviation helps filter outliers.

With this method, we select the top 20 challenging tasks for GPT-4, and we find the scenarios are commonly challenging for other models as well (compare Figure G.4 and G.5). We use SOTOPIA-hard to refer to these 20 challenging tasks.

We run two experiments: (1) humans interact with GPT-4, and (2) humans interact with each other, both under the SOTOPIA-hard setting. We collect 20 human-human interactions and 40 human-GPT-4 interactions covering all 20 tasks in SOTOPIA-hard. Note that humans are not aware of the identity of their partners during the interaction.¹¹

We then evaluate humans and GPT-4’s interactions with GPT-4 and human annotators as the evaluators. As shown in Table 3, humans perform significantly better than GPT-4 in the GOAL dimension.

It is also worth noting that humans on average produce 16.8 words per turn, while GPT-4 produces 45.5 words per turn, which indicates humans are more efficient in social interactions.

Specifically, we find that GPT-4 always rephrases the utterance back at the other agent and then answers, which is a communication skill called active listening (Harry Weger & Robinson, 2014), whereas humans typically directly answer. This is likely due to the fact that GPT-4 is trained with a large amount of human feedback, which makes it overly helpful in the conversation.

Qualitatively, Humans are usually more strategic than GPT-4 during interaction. When bargaining, if the GPT-4 agent has a buying target set at \$454, it sometimes starts its bid at that exact price (Figure H.10). Consequently, any subsequent negotiations push the final agreed price above its initial target. In contrast, human annotators (e.g. Figure H.11) begin the negotiation at a lower bid of \$400, and often reaches an agreement with the seller at a price that’s still below the GPT-4’s target. Humans are also more persistent in their goals. When trying to settle one a music to listen to, the model tends to propose a compromised solution (e.g. Figure H.12), such as each one listening to a few selected songs. Humans, however, tend to persist in adhering to their set goals (e.g. Figure H.13).

	GOAL	BEL	REL	KNO	SEC	SOC	FIN
GPT-4 (w H)	4.85	9.25	0.70	2.80	0	0	0.50
Human (w G)	5.95*	9.15	0.60	2.95	0	-0.60	0.70
Human (w H)	6.15*	9.10	0.80	2.65	0	-0.10	0.45

Table 3: Human and GPT-4 performance on different dimensions on SOTOPIA-hard. SOC and SEC have the scale of -10 to 0, REL and FIN have the scale of -5 to 5, and others have the scale of 0 to 10. (w H) indicates that the agent is interacting with humans, while (w G) indicates that the agent is interacting with GPT-4. * indicates the difference is significant compared to GPT-4 (w H) with $p < 0.05$ under student’s t-test. We also report the agents performance evaluated by human annotators (Table G.4), which shows similar trends.

¹¹See Appendix F for the detailed instructions and the chat interface.

8 RELATED WORK

Enabling artificial agents to interact with each other and with humans has been studied in different fields. Our work draws inspiration from literature on social intelligence, dialogue systems, and simulations of social interactions. See Appendix A for an extended discussion.

Static social intelligence benchmarks To evaluate social intelligence in AI systems, researchers have proposed a variety of static benchmarks. Some of them are inspired by clinical tests of social intelligence for humans, such as the ToMi dataset (Le et al., 2019) and the FauxPas dataset (Shapira et al., 2023b). Other benchmarks are designed to evaluate social intelligence in the context of social commonsense reasoning, such as SocialQA (Sap et al., 2019) and SocialIQ (Zadeh et al., 2019a). With the rapid development of LLMs, some of the benchmarks gradually become saturated. Recent works synthesize existing benchmarks and propose new adversarial datasets to evaluate social intelligence (Shapira et al., 2023a; Wilf et al., 2023). Although these benchmarks are harder than their predecessors, they still lack the dynamic nature of social interactions and the rich social context, which is deemed insufficient for evaluating social intelligence in AI systems (Lee et al., 2023).

Task-oriented and open-domain dialogue systems Dialogue systems offer a natural interface to interact with AI systems. Task-oriented dialogue systems are designed to help users accomplish specific tasks, often evaluated with task success rate or user satisfaction (Hosseini-Asl et al., 2022; FAIR et al., 2022; Wang et al., 2019) without generalizing to other tasks.¹² Open-domain dialogue systems are designed to have “chit-chat” with users (Kann et al., 2022; Kim et al., 2023), often incorporate personal information to make conversations more engaging (Zhang et al., 2018a; Liu et al., 2020; Baha et al., 2023; Doğruöz & Skantze, 2021; Skantze & Doğruöz, 2023). Such systems often appear to understand the subjects deeper than they actually do without a specific goal during the interaction (Weizenbaum, 1966, Eliza effect). SOTOPIA forces agents to maintain their social persona and achieve *explicit* social goals spontaneously, which is more challenging than the existing dialogue systems.

Simulations of social interactions with LLMs LLMs contain a large amount of knowledge about the world and can generate human-like responses based on the social context (Park et al., 2023; Kim et al., 2023; West et al., 2022). Recently, researchers have used LLMs to simulate social interactions for various purposes, such as facilitating the design of social media platform (Park et al., 2022), producing believable proxies of human behaviors (Park et al., 2023), and developing software collaboratively (Qian et al., 2023). However, these works focus on showcasing the capabilities of LLMs in simulating social interactions rather than systematic evaluation of agents’ social interactions. Specifically, Park et al. (2023) use TrueSkill rating to evaluate agents’ performance in aspects such as memorization, planning, and reflecting the past actions while ignoring other important dimensions such as SOC and SEC during social interactions. CAMEL Li et al. (2023a) simulates the collaboration task solving process in LLMs, Gentopia Xu et al. (2023) works on augmented LLMs with tools to facilitate collaboration, while ChatDev Qian et al. (2023) focuses on the software development domain.

Multi-agent coordination Although in paper we focus on evaluating language agents, our research is heavily-inspired by recent advances in multi-agent coordination and social learning Lowe et al. (2017); McKee et al. (2020); Hu et al. (2020); Zhu et al. (2021); Liu et al. (2022); Trivedi et al. (2023). Our setting is more realistic than the commonly-used assumptions that agents have either zero (other-play) or extensive knowledge of each other’s policies (self-play).

9 CONCLUSION

In this paper, we present SOTOPIA, an environment that can be used to simulate the goal-driven social interactions of agents in a variety of social scenarios. Different from most previous benchmarks for social intelligence, SOTOPIA is interactive, goal-oriented, and covers a large range of realistic social tasks. Our experiments demonstrate that GPT-4 could automate the evaluation of agent performance based on SOTOPIA-EVAL. Building on this, we show that SOTOPIA can be used for understanding not only the differences among models but also the difference between models and humans in terms of social interaction abilities. We discuss the limitations of SOTOPIA and future directions in Appendix B. Our findings indicate that SOTOPIA has potential as a platform for assessing and enhancing the social skills of language-based agents.

¹²Here, we consider a broader concept of task-oriented dialogue systems including action-taking abilities.